

## Unit

1

Lesson  
3

# Spark The Challenge!

## Shape It Up in 3D!

Time Frame: [90 min]

### Lesson Objectives:

**By the end of this lesson students will be able to:**

- Identify and describe the basic tools in Fusion 360, such as Sketch, Extrude, Revolve, and Sweep
- Design components of a Smart Walker (like a base frame and handle).
- Visualize your design idea digitally, applying what they've learned about accessibility and user needs

## Challenge questions of the lesson

- Why do you think digital modeling is important before building a physical prototype?
- If your Smart Walker is meant for someone with shaky hands, how might that affect your design?
- How can designing digitally in Fusion 360 help you spot problems before they happen?
- What would you change about your walker design if it had to work outdoors, on rough surfaces?

## SEL Relation

- **Self-Awareness:** Students reflect on how they learn new tools and overcome tech challenges.
- **Social Awareness:** Empathy mapping encourages understanding of users with mobility challenges.
- **Responsible Decision-Making:** Students make design choices based on user needs and constraints.
- **Relationship Skills:** Working in pairs or teams to discuss ideas, provide feedback, and share screens.
- **Self-Management:** Staying focused during Fusion 360 tasks and managing frustration during mistakes or software errors.

## Engage



### Teacher's Role:

- **Questioner:** Ask Thought-Provoking Questions:
  - Prompt curiosity by asking open-ended questions such as:
 

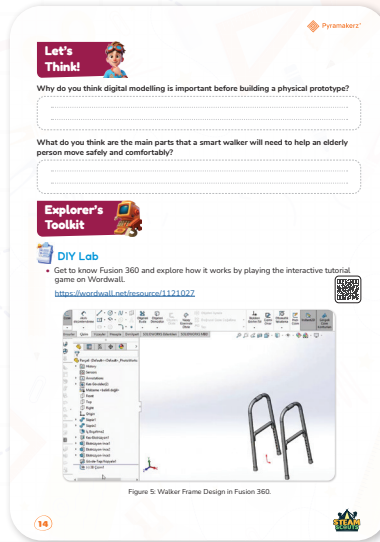
“What do you think engineers use to design things like cars, phones, or wheelchairs?”

“Have you ever designed something before—even in a game like Minecraft or Roblox?”
  - Spark student thinking and set the stage for meaningful discussion.
- **Connector:** Make Real-World Connections
  - Relate engineering and design tools to familiar experiences and interests (e.g., video games, home gadgets, building toys).
  - Use real objects, quick demos, or relatable analogies to make abstract concepts concrete and engaging.
- **Encourager of Student Voice:** Foster Inclusive Participation
  - Invite all students to share their ideas, reminding them there are no wrong answers in brainstorming.
  - Use participation structures like Think–Pair–Share, sticky-note brainstorming, or quick sketches to ensure every voice is heard.
- **Safe Space Creator:** Establish a Judgment-Free Zone
  - Reinforce that early design is about exploring, not getting the “right” answer.
  - Celebrate curiosity and creativity over perfection—encouraging risk-taking and experimentation.

### Expected From Students:

- **Sharing Ideas Freely:**
  - Offering guesses, prior knowledge, or even “I wonder...” thoughts
  - Participating in pair or group discussions
- **Making Real-World Connections:**
  - Relating engineering and design to their own lives (e.g., games, inventions, daily tools)
- **Asking Questions:**
  - Starting to wonder: “Can I make a real walker design using this software?”
- **Getting Curious:**
  - Feeling excited or intrigued about what’s coming next
  - Thinking about problems they could solve through design

## Parts of the book included.



### Let's Think!



#### Objective:

Students will reflect on the importance of digital modeling in the engineering design process and identify key functional components needed in a Smart Walker to support elderly mobility, safety, and comfort.

#### Possible Answers:

##### Why is digital modelling important before building a physical prototype?

Digital modelling helps us test ideas safely, spot design problems early, and make improvements before using real materials. It saves time, reduces cost, and allows easy changes before building the physical prototype.

##### What are the main parts a smart walker needs to help an elderly person move safely and comfortably?

A smart walker needs a strong frame, comfortable handles, stable wheels, brakes for safety, sensors to detect obstacles, and smart features like alerts or lights to support balance, safety, and ease of use.

#### Possible Strategies for Implementation:

##### Strategy 1: Think–Pair–Share

###### • Implementation:

- Ask the question aloud:  
“What do you think engineers use to design things like cars, phones, or wheelchairs?”
- Give students 1 minute to think silently.
- Have them pair up and discuss their ideas (1–2 minutes).
- Then, share out answers with the class.



## Strategy 2: Digital Design Connection

- **Implementation:**
  - Ask: “Who’s built something in Minecraft, Roblox, or a game like The Sims?”
  - Then say: “Guess what—engineers use software kind of like that to build real things! Cars, wheelchairs, even buildings. It’s called CAD software.”
  - Purpose:  
Connects their digital play experience to real-world engineering tools like Fusion 360.

## Strategy 3: Think & Predict

- **Implementation:**
  - Ask: “If you had to build something totally new from scratch, what could go wrong if you just started building without designing first?”
  - Write down 2 possible problems they might face
  - Share with a partner or in a quick classroom discussion

## Explorer’s Toolkit



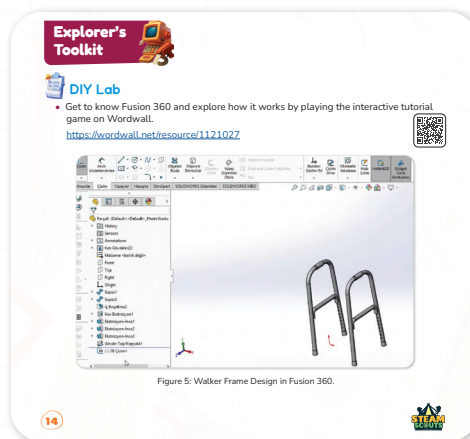
## Teacher's Role:

- **Empathy Facilitator:** Guide students through the first interactive game or simulation, prompting them to think deeply about the experiences of elderly individuals. Encourage discussion, reflection, and emotional connection to develop empathy-driven design thinking.
- **Demonstrator of Design Tools:** Model each step of the digital design process in Fusion 360. Clearly show how to begin a 2D sketch, select the XY plane, and use the interface confidently.
- **Supporter of Skill Building:** Assist students with workspace setup in Fusion 360, including naming components, navigating menus, and using core sketching tools. Offer one-on-one or small-group support as needed.
- **CAD Coach:** Demonstrate foundational CAD practices, including drawing with rectangles and lines, applying constraints, and using dimensions to define precise geometry. Encourage students to explore while reinforcing accuracy and best practices.
- **Organizer and Encourager:** Promote effective design habits by teaching students how to break complex models into separate components for easier organization, editing, and collaboration. Encourage students to stay organized and iterate thoughtfully.

## Expected From Students:

- **Sketching the object thoughtfully**
  - Drawing the chair from observation
  - Labeling parts like legs, seat, and backrest
- **Breaking down the object**
  - Identifying how many parts it has
  - Asking: What parts would need to be made separately?
- **Translating physical to digital**
  - Noting which geometric shapes make up each part
  - Predicting what Fusion 360 tools they might use to recreate it
- **Collaborating and sharing ideas**
  - Discussing with peers: “How would you design the backrest?”  
“Would you use a fillet or loft tool here?”

## Parts of the book included.



### DIY Lab

#### Objective:

- Students will explore the Fusion 360 interface and understand basic CAD functions by engaging with an interactive tutorial game. This will build foundational skills for using digital modeling tools in future design tasks.
- Understand and Apply Core CAD Functions: Students will be able to identify and use the basic tools in Fusion 360 (2D Sketch, Extrude, Fillet, Shell, Joint, Offset Plane) to design and modify simple 3D models, demonstrating foundational skills in computer-aided design (CAD).

#### Materials Needed for activity one:

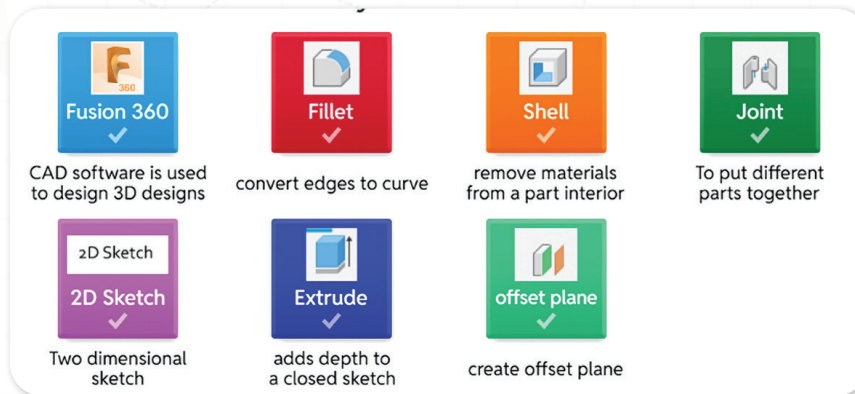
- Internet-connected devices (laptop or desktop)
- Access to the Wordwall game: <https://wordwall.net/resource/1121027>
- Fusion 360 installed and launched on student devices



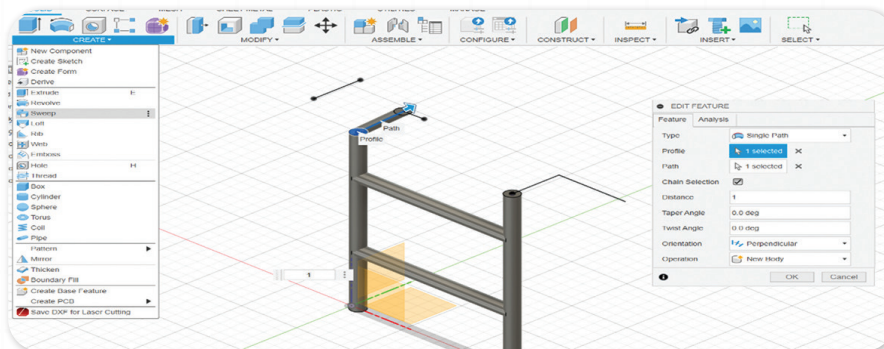
## Materials Needed for activity two:

- Laptop or desktop computer with Fusion 360 installed
- Internet access (for saving to the cloud and teacher support, if applicable)
- Student Fusion 360 accounts (set up prior to the activity)
- Mouse (recommended for more precise sketching)

## Possible answers for the wordwall activity (First activity):



## Answer for Fusion activity (Activity 2):



## Activity Instructions for activity one:

### 1. Launch the Wordwall Game:

Direct students to open the Fusion 360 tutorial game using the link provided.

### 2. Play & Explore:

Allow students time to complete the interactive tutorial. Encourage them to read carefully and test features in Fusion 360 as they go.

### 3. Observe & Record:

Ask students to note down at least 3 new things they learned about the Fusion 360 interface or tools.

### 4. Class Discussion (Optional): After the game, discuss as a class:

- “What surprised you?”
- “What tools or steps do you want to try next?”

## Activity Instructions for activity two:

Second activity resources: [https://www.youtube.com/watch?v=5jMmT-ATlFo&list=PL0sBSJdx2cgHuTh\\_t0yBinuWz\\_BFqlocy&index=7](https://www.youtube.com/watch?v=5jMmT-ATlFo&list=PL0sBSJdx2cgHuTh_t0yBinuWz_BFqlocy&index=7)

[https://drive.google.com/file/d/1ELLHiyDlC4bKkdT4FOnxdmzuLE9DqHn/view?usp=drive\\_link](https://drive.google.com/file/d/1ELLHiyDlC4bKkdT4FOnxdmzuLE9DqHn/view?usp=drive_link)



### 1. Review the Challenge:

Remind students that they are now beginning the digital modeling phase of their Smart Walker design, focusing first on creating an organized structure.

### 2. Organize the Design:

- Instruct students to break their walker into logical components (e.g., frame, handles, legs, wheels).
- Explain that separating parts will help with editing, scaling, and future upgrades.

### 3. Launch Fusion 360:

- Ensure Fusion 360 is open and ready.
- Guide students to create a new project or open their existing Smart Walker file.

### 4. Create a New Sketch:

- Instruct students to start a 2D sketch on the XY plane.
- Emphasize correct naming of the sketch (e.g., "Walker Base") for organization.

### 5. Sketch the Base Frame:

- Use basic shapes like rectangles, lines, and circles to outline the walker's base.
- Remind them to include structural parts like legs and support bars.

### 6. Apply Constraints and Dimensions: (optional at this stage)

- If appropriate for your learners, have students apply constraints (e.g., horizontal, equal) and add dimensions to define size and alignment.

### 7. Save and Review:

- Students should save their work and share it with a partner or teacher for quick feedback before moving on.

## Possible Strategies for Implementation:

### Strategy 1: "Think Like a CAD Designer"

Students sketch a chair on paper while answering guided questions.

#### • Guiding Questions:

1. "How many separate parts make up this chair?"
  - \*(Example: Seat, 4 legs, backrest, screws/glue joints.) \*
2. "What basic 3D shapes do you see?"
  - (Rectangles, cylinders, curved surfaces.)



3. "Which parts would need to be separate in Fusion 360?"

- o \*(Legs must be individual for adjustments; seat/backrest could be one piece.) \*

### Strategy 2: Shape Recognition & Translation

#### Implementation:

Ask students to circle or highlight the basic shapes they see in their sketch (e.g., rectangles, cylinders).

- "What shapes do you see?"
- "Which of these could be sketched in Fusion 360?"
- "What tool would you use? (Sketch? Extrude? Fillet?)"

#### You could model one example:

"I'd start with the seat—it's a rectangle, so I'd sketch a rectangle, then extrude it into a box."

### Strategy 3: Think-Pair-Share

#### How to implement:

Students first think individually about where they might use each Fusion 360 tool (e.g., Fillet, Shell) in a real-world object. They then pair up to discuss their ideas. Finally, pairs share their thoughts with the class, promoting participation and deeper understanding.

### Strategy4: Round Robin

#### How to implement:

In small groups, students take turns naming a Fusion 360 tool and describing its function or giving an example of its use. This ensures every student contributes and reinforces tool definitions.

### Strategy5: Quiz-Quiz-Trade

#### How to implement:

- Each student receives a card with a tool name or function.
- They quiz a partner, trade cards, and repeat with new partners.
- This activity helps reinforce vocabulary and tool recognition in a fun, interactive way.

### Strategy6: See-Think-Wonder

#### How to implement:

- Show the tool icons and descriptions.
- Students write or discuss:
  - What do you see? (e.g., "I see a blue box labeled 'Extrude.'")
  - What do you think this tool does?
  - What do you wonder about how or why this tool is used in design?

### For the second activity:

#### Strategy1: Kagan Strategy: Think-Pair-Share

##### How to implement:

- Students think about why components help organize designs, pair up to discuss, then share with class.

#### Strategy2: See-Think-Wonder

##### How to implement:

- Show the Fusion 360 interface with components and sketches. Ask:
  - What do you see? (e.g., “I see components named Frame and Wheels.”)
  - What do you think is the purpose of components?
  - What do you wonder about how components help in design?

#### Strategy3: Step-by-Step Modeling

##### How to implement:

- Break down the process: first create components, then sketch base frame, then extrude and add details.

#### Strategy4: Use Video Tutorials

##### How to implement:

- Supplement with beginner Fusion 360 videos covering components and sketching basics.

## Explain



### Teacher's Role:

#### • Introduce Key Concepts Clearly

- Introduce formal terms like CAD, Sketch, Extrude, Component, Join, Fillet, etc.
- Relate them directly to the chair parts or Smart Walker ideas students explored earlier.

“Remember how you sketched the chair’s legs as cylinders? In Fusion 360, we’ll sketch a circle, then extrude it to make it 3D.”

#### • Demonstrate Tools Step-by-Step

- Project Fusion 360 on the screen and walk through creating a simple part (like the seat or handle of a walker).
- Explain what each tool does and why it’s used.

#### • Connect to Real Engineering

- Show how engineers use these tools in industries like product design, healthcare, or robotics.



- **Encourage Student Questions**

- Pause and ask: “Why do you think we’re using the ‘Join’ tool here?”
- “What might happen if our parts aren’t aligned correctly?”

## Expected from Students:

- **Listen actively**

- Focus on understanding why and how each CAD tool is used.
- Take notes or label diagrams with new vocabulary.

- **Connect concepts to earlier exploration**

- Recognize how their chair sketch or walker idea could now be modeled using the tools being demonstrated.

- **Ask Clarifying Questions**

- “Can we use the mirror tool for symmetrical parts?”
- “What if I want to change the size later?”

- **Practice or follow along if possible**

- If devices are available, students should replicate the demo in Fusion 360 while the teacher models.

## Parts of the book included.



### Screen Lessons

Click on the link to learn more about how fusion work.

<https://www.youtube.com/watch?v=ZYXSJ6EQ9S4>



## Screen Lessons

### Objective:

Students will explore how to use Autodesk Fusion 360 to design and model 3D objects. By watching the video.

Resources: <https://www.youtube.com/watch?v=zy-PS-c16ml>

<https://www.youtube.com/watch?v=ZYXSJ6EQ9S4> (extra video to create components)



## Possible Strategies for implementation:

### 1. Pre-Watching Strategies

**Strategy:** Quick Connect + Prediction

**How to implement:**

- Write on the board: “What do you think CAD means? Why would engineers use computers to design instead of drawing on paper?”
- Allow students to:
  - Share guesses (individually, with partners, or on sticky notes)
  - Make a prediction: “What kinds of things do you think CAD is used to design?”



## 2. During Watching: Active Strategies

**Strategy:** Guided Viewing with a Simple Note Sheet

**How to implement:**

- Provide a 3-question viewing guide (printed or on a slide):
  - What is CAD?
  - What kinds of things are designed using CAD?
  - What are two cool facts or examples mentioned in the video?

**Engagement Tip:**

- Pause the video at key points to:
  - Highlight terminology (like "3D modeling", "prototype", "engineer")
  - Ask short verbal check-ins:  
"What tool are they using here?"  
"Why is a digital design helpful?"

## 3. Post-Watching Strategies

**Strategy:** Pair-Share + Group Reflection

**How to implement:**

- Ask students to pair up and answer:
  - "How is CAD like building with LEGO or Minecraft?"
  - "What part of a Smart Walker do you think we could design in CAD?"
- As a class, build a mind map on the board:
  - Center: CAD
  - Branches: Tools used, real-world applications, benefits, connections to Fusion 360

## Elaborate and Evaluate



## Teacher's Role:

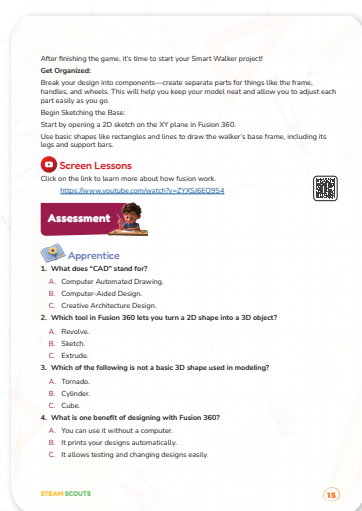
- **Connect Prior Learning to New Applications:**
  - Help students link CAD tools (Sketch, Extrude, Join, etc.) to the features of their Smart Walker.
  - Revisit ideas from the "Explore" and "Explain" phases and ask: "Which tools can you use to model the wheels, handles, or support frame?"
- **Encourage Deeper Design Thinking:**
  - Prompt students to make design choices based on user needs:  
"How can your walker be more comfortable or safer for someone with arthritis?"

- **Support Iteration & Customization:**
  - Allow students to revise their designs as they build in Fusion 360.
  - Offer mini-feedback moments or peer reviews.
- **Promote Student Ownership:**
  - Give students room to innovate and personalize their walkers: “Can you add one smart feature that hasn’t been done before?”
- **Circulate, Support, and Ask “Why”:**
  - Walk around, ask guiding questions: “Why did you choose that shape?” “What problem is this solving for the user?”

## Expected From Students:

- **Designing Their Smart Walker or a Key Component:**
  - Using Fusion 360 tools to model at least 2 parts.
  - Combining parts into an assembled digital model.
- **Making User-Centered Design Decisions:**
  - Choosing dimensions, features, and materials based on the user persona (e.g., elderly person, someone with mobility issues)
- **Using CAD Vocabulary While Explaining Choices:**
  - Explaining their designs using terms like “extrude,” “join,” “component,” “mirror,” etc.
- **Iterating & Refining Their Design**
  - Making edits to improve comfort, safety, or ease of use.
  - Testing whether parts align or fit logically in the model.
- **Collaborating & Reflecting:**
  - Sharing progress with peers, offering and receiving constructive feedback.
  - Reflecting on what worked and what could improve.

## Parts of the book included.





## Apprentice

### Objective:

Students will identify and describe key concepts of Computer-Aided Design (CAD), recognize common tools and functions in Fusion 360, and distinguish between basic and non-basic 3D shapes used in digital modeling. They will also understand the benefits of using digital modeling software in the engineering design process.

### Possible Answers:

**Assessment**

**Apprentice**

1. What does "CAD" stand for?  
A. Computer Automated Drawing.  
B. Computer-Aided Design.  
C. Creative Architecture Design.
2. Which tool in Fusion 360 lets you turn a 2D shape into a 3D object?  
A. Revolve.  
B. Sketch.  
C. Extrude.
3. Which of the following is not a basic 3D shape used in modeling?  
A. Tornado.  
B. Cylinder.  
C. Cube.
4. What is one benefit of designing with Fusion 360?  
A. You can use it without a computer.  
B. It prints your designs automatically.  
C. It allows testing and changing designs easily.

## Possible Strategies for Implementation:

### 1. Think–Pair–Share (Before the Quiz)

#### Implementation:

- Before students complete the assessment, ask them to briefly discuss questions like “What does CAD mean to you?” or “What tools in Fusion 360 have you used so far?”
- Students think individually, discuss with a partner, and share highlights with the class to activate prior knowledge.

### 2. Interactive Quiz Game (Digital or Physical)

#### Implementation:

- Turn the multiple-choice questions into a live quiz using tools like Kahoot, Quizizz, or Wordwall.
- This boosts engagement and provides instant feedback while reinforcing concepts.



## Builder

### Objective:

Students will begin developing a digital prototype of their Smart Walker by organizing the design into functional components and creating a 2D sketch of the base frame in Fusion 360 using basic geometric shapes.

### Materials Needed:

- Computers or laptops with Fusion 360 installed
- Student Fusion 360 accounts (set up prior to activity)

- Mouse (recommended for more precise design control)
- Internet access (if working with cloud-based saving or tutorials)

### Helping Resources:

[https://www.youtube.com/playlist?list=PLOsBSJdx2cgHuTh\\_t0yBinuWz\\_BFqlocy](https://www.youtube.com/playlist?list=PLOsBSJdx2cgHuTh_t0yBinuWz_BFqlocy)



(This is a playlist consisting of 8 videos showing how to design all the parts of a walking stick in fusion 360)

We used the first 2 videos only, High achievers can make more tasks from the rest of the videos



[https://www.youtube.com/watch?v=PtGCP1qFMe4&list=PLOsBSJdx2cgHuTh\\_t0yBinuWz\\_BFqlocy&index=8](https://www.youtube.com/watch?v=PtGCP1qFMe4&list=PLOsBSJdx2cgHuTh_t0yBinuWz_BFqlocy&index=8)

[https://www.youtube.com/watch?v=5jMmT-ATlFo&list=PLOsBSJdx2cgHuTh\\_t0yBinuWz\\_BFqlocy&index=7](https://www.youtube.com/watch?v=5jMmT-ATlFo&list=PLOsBSJdx2cgHuTh_t0yBinuWz_BFqlocy&index=7)



### Activity Instructions:

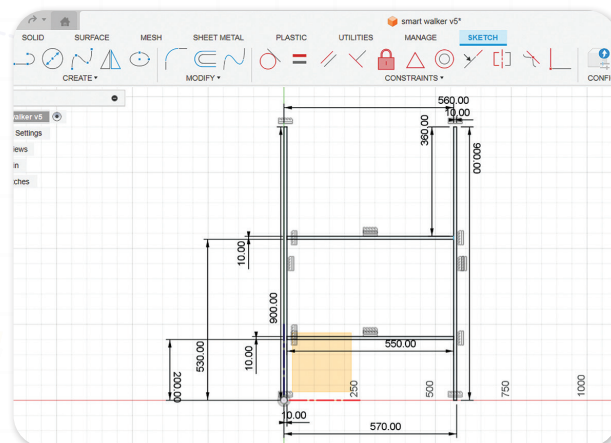
#### 1. First, search for the dimensions of a walker



- Rule no. 1, name the fusion file and creating a new component, name it. Then press on create sketch and sketch on the front plane

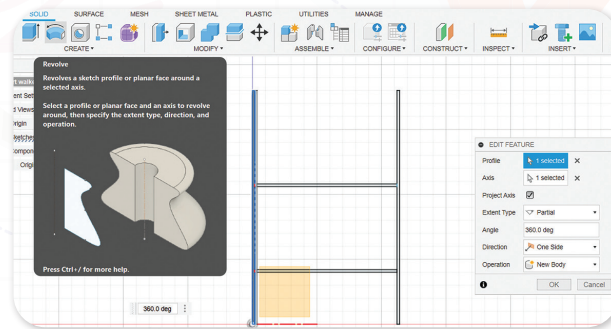
#### 2. creating front side of smart walker

- press on create → rectangle → 2 points rectangle and draw the front of the walker (all rectangle shapes) with the dimensions from the picture

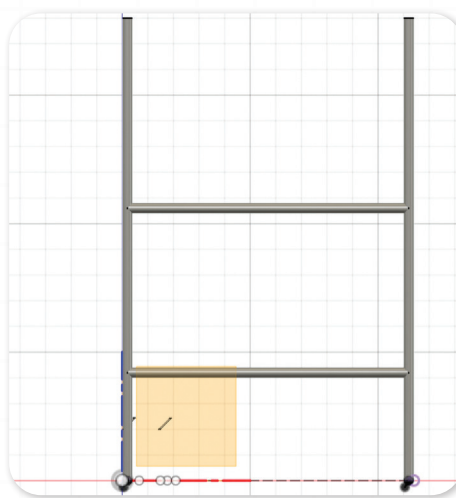




- Press finish sketch, press Revolve → choose 1 profiles (1 rectangle), axis (inner line), 360 degree, new body

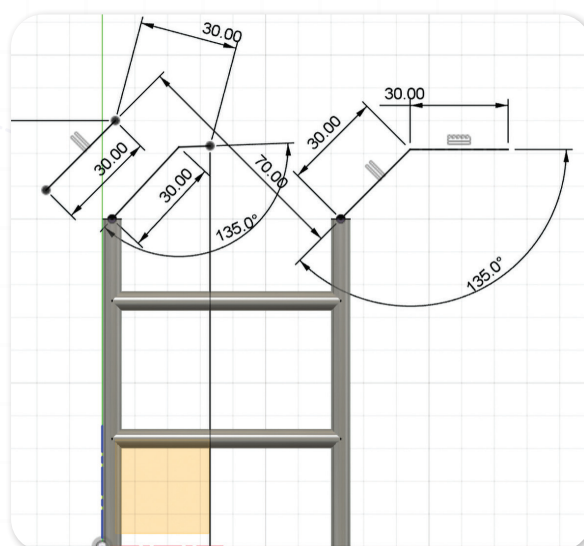


- Repeat the last step for the other three rectangles, now you have the front side of your walker ready

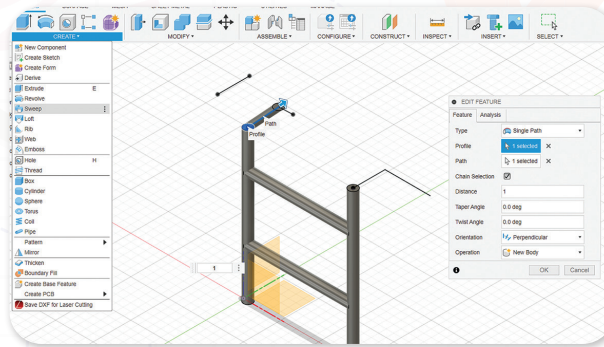


### 3. Creating the handle using the sweep tool

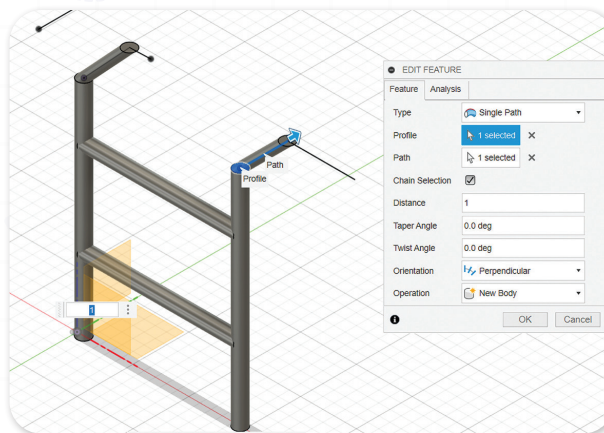
- Create a new sketch on front plane also, and then show the first sketch of the front side and from mid-point draw a line with length 30 mm
- Press on coincident and connect 2 points together
- Then draw another line with angle 135 degree and length 30 mm
- Repeat for the other rod



- Press on solid → create → then sweep, choose the profile (circle at the edge of cylinder), path (the line), type: single path, operation: new body

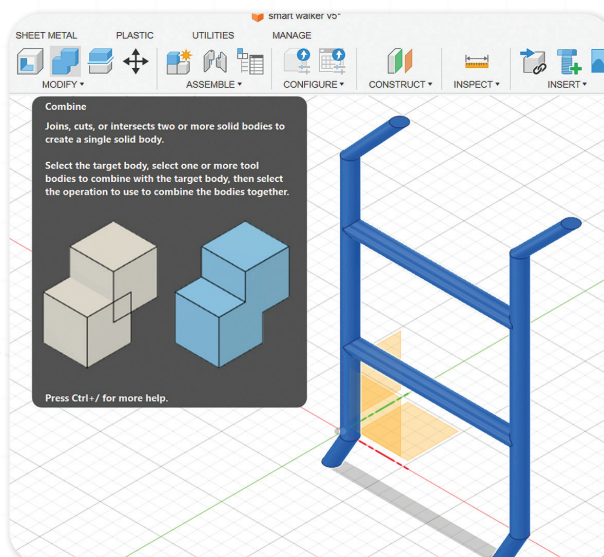


- Repeat the last step for the other rod, now you have the handle ready

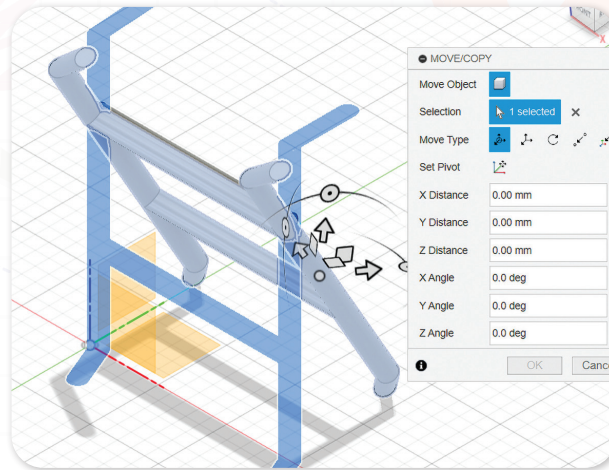


- From select → window selection, select the whole design
- From modify → press combine

Now you have the whole front side of the walker as 1 body

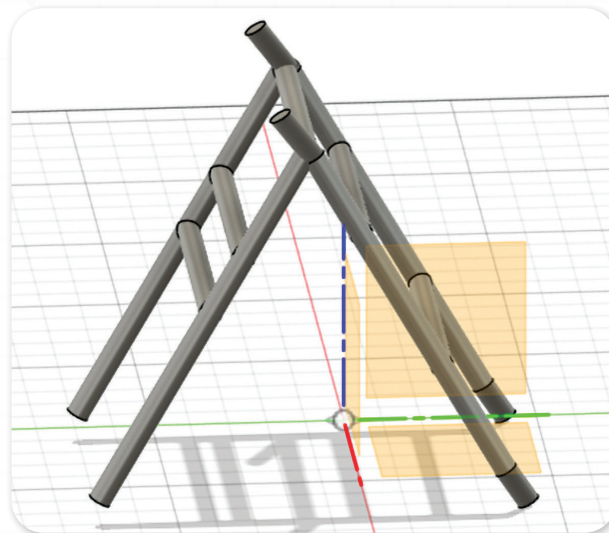


- Press M (move) and rotate from x angle 45 degrees

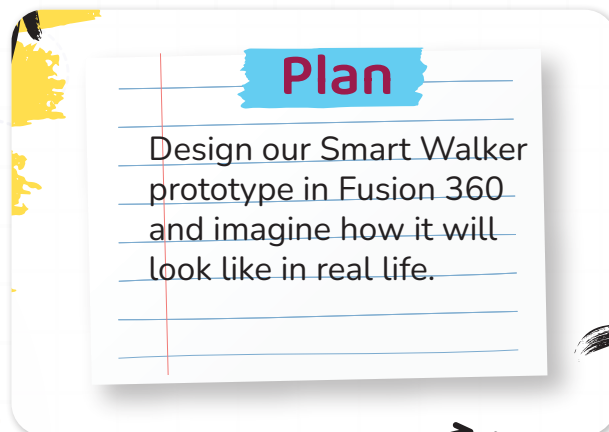


- Repeat the past steps (without the steps of the handle) to make a similar design to the front side which will be the back side of the walker and also press combine to have it as one body
- Then press M (move) and rotate it from x angle (- 45) degrees

Now, your smart walker design is ready



Don't forget to go back to the EDP page and add these to planning part.





## Now I Can

### Objective:

Students will reflect on their learning experience with Fusion 360, identifying and describing key tools such as Sketch, Extrude, Revolve, and Sweep. They will also assess their ability to design parts for the Smart Walker and consider accessibility and user needs in their digital designs. This self-reflection will help students evaluate their progress and develop a deeper understanding of the design process.

### Activity Instructions:

#### Step 1: Revisit the Lesson

Students review their digital modeling work, component sketches, and design notes in Fusion 360.

#### They reflect on:

- What is the purpose of designing a Smart Walker?
- Which tools in Fusion 360 did I use (e.g., Sketch, Extrude, Revolve, Sweep), and how did I use them?
- What components did I design, and how do they help someone with mobility challenges?
- How did I use digital tools to plan before building anything physical?

#### Step 2: Student Self-Reflection

Students complete the sentence starters or journal prompts:

- “Now I can explain why Smart Walkers are important for real users.”
- “I used \_\_\_ and \_\_\_ in Fusion 360 to bring my design idea to life.”
- “One component I created is the \_\_\_, and its job is to \_\_\_.”
- “Sketching in Fusion helped me \_\_\_ before I built anything physically.”
- “One way I made my design more accessible was by...”

Encourage them to include visuals, sketches, or screenshots if using a digital format.

#### Step 3: Small Group Discussion

In small groups, students share one thing they feel proud of from their design and one challenge they faced.

- Use Think–Pair–Share or Inside–Outside Circle with prompts like:
  - “What tool in Fusion 360 helped you the most—and why?”
  - “Which part of your Smart Walker design is most helpful to the user?”
  - “How did digital modeling help you solve a real-world problem?”



## Relation to Standards

### 1. ISTE Standards for Students (International Society for Technology in Education)

- **1. Innovative Designer:** Students use a variety of technologies within a design process to identify and solve problems by creating new, useful or imaginative solutions.
- **4. Computational Thinker:** Students develop and employ strategies for understanding and solving problems in ways that leverage the power of technological methods.
- **6. Creative Communicator**

### 2. Next Generation Science Standards (NGSS) – Engineering Design (Middle School)

- **MS-ETS1-1:** Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution.
- **MS-ETS1-2:** Evaluate competing design solutions using a systematic process to determine how well they meet the criteria and constraints of the problem.
- **MS-ETS1-4:** Develop a model to generate data for iterative testing and modification of a proposed object, tool, or process.

### 3. Common Core State Standards (CCSS) – Math & ELA Integration

- **CCSS.MATH.CONTENT.7.G.A.2:** Draw (freehand, with ruler and protractor, and with technology) geometric shapes with given conditions..
- **CCSS.ELA-LITERACY.SL.7.4:** Present claims and findings, emphasizing salient points in a focused, coherent manner.

### 4. 21st Century Skills (P21 Framework) & Career Readiness

- **Critical Thinking & Problem Solving**
  - **Creativity & Innovation:** Encouraged through personalization, feature development, and smart tech integration.
  - **Technology Literacy:** Direct use of professional 3D modeling software (Fusion 360).
  - **Career Awareness**